total entropy in the bar between 10 p.m. and 10:30 p.m. in the two scenarios—one in which you trust your memories, and the other in which things spontaneously arrange themselves from an initial state of disorder to be consistent with what you see, now, at 10:30 p.m.—there is an enormous entropy difference. The latter scenario, every step of the way, has *hugely* more entropy than the former scenario, and so, from the standpoint of probability, is hugely more likely.

14 A closely related point is that should we convince ourselves that the world we see right now just coalesced out of total disorder, the exact same reasoning—invoked anytime later—would require us to abandon our current belief and, instead, attribute the ordered world to a yet more recent fluctuation. Thus, in this way of thinking, every next moment invalidates the beliefs held in each previous moment, a distinctly unconvincing way of explaining the cosmos.

15 That is, a black hole of a given size contains more entropy than *anything* else of the same size.

16 Even though Feynman's sum over histories approach might seem to make the particle aspect prominent, it is just a particular interpretation of probability *waves* (since it involves many histories for a single particle, each making its own probabilistic contribution), and so is subsumed by the wavelike side of complementarity. When we speak of something behaving like a particle, we will always mean a conventional particle that travels along one and only one trajectory.

17 If you find this section tough going, you can safely move on to the next section without loss of continuity. But I encourage you to try to get through it, as the results are truly stupendous.

18 Quantum mechanics, rightly, has a reputation as being anything but smooth and gradual; rather, as we will see explicitly in later chapters, it reveals a turbulent and jittery microcosmos. The origin of this jitteriness is the probabilistic nature of the wavefunction—even though things can be one way at one moment, there is a probability that they will be significantly different a moment later—*not* an ever-present jittery quality of the wavefunction itself.